

MTH209: Assignment 3

Colored Dog Compression

There is a colored image of a dog in your repository. The number of pixels are 250×250 . Implement image compression in the following way:

1. Stack the three channels of colors on top of each other to create a 750×250 matrix
2. Implement SVD with $k = 50$ components.
3. Unstack the lower-dimensional matrix back to a three channel array. This can be done by creating an empty array using the code below and then filling the appropriate components.

```
newdog <- array(0,dim = c(250,250,3))
```

4. Save this unstacked image in an array called `newdog`. This array should have the same dimensions as above.

INSTRUCTIONS: copy and paste all the relevant code in the `assignment3.R` file in the GitHub repository. NO PARTIAL MARKING for this assignment. DO NOT PRINT UNNECESARY THINGS IN THE CONSOLE!